Does Trade Raise Welfare for Every Canadian?

AI-enhanced project brief – Option B

Work in pairs • Choose one option • 100 points • Deadline and submission link: see Canvas

Names _____ **Section** _____

Your decision question. Advise a non-partisan Canadian distributional review. Define welfare and the population whose welfare counts. Does the evidence support higher average welfare? Does it support the stronger claim that every Canadian gains? Your common foundation must combine labour income or employment with consumption costs for all five income quintiles. You may add one or two measurable non-income dimensions. Build at least four household profiles, identify plausible gains and losses, and recommend one targeted adjustment policy.

1. Context and goal

Trade can affect a household through several channels. Export opportunities, import competition, productivity, and labour reallocation can change employment and earnings. Imports and competition can change the prices and varieties in the household's consumption basket. A household can gain as a consumer while losing as a worker, or the reverse.

Canadian evidence makes the tension concrete. Historical liberalization offers evidence on productivity, plant adjustment, employment, and earnings. The 2025–26 protectionist episode offers recent evidence on trade exposure and retail-price pass-through. Neither episode, by itself, measures the total gain from all Canadian trade. Your task is to combine the two channels carefully, state what is observed and what is a scenario, and distinguish an aggregate average from a universal claim about every person.

The wage–employment and consumption-price ledger is the common analytical foundation, not a complete definition of welfare. You may add one or two measurable dimensions such as job security, hours and leisure, product variety or access, health, housing and commuting, environmental quality, or regional resilience. For each addition, name whose welfare it measures, why it matters, its unit and preferred direction, trade mechanism, time horizon, and open data. Report dimensions separately unless you can defend the weights used to combine them; do not manufacture an arbitrary welfare index. Put other omitted channels in the limitations.

2. Required output – one package per pair

1. **One PDF** containing an exactly four-page narrative memo, followed by a references/data appendix of at most one page and a reproducibility/AI-use appendix of at most one page. Use 11-point type, ordinary margins, and no cover page.
2. **One research ZIP** containing a README, unchanged raw data, processed data, code, generated figures, a source audit, verification report, AI log, and partner contribution statement.
3. **Two principal exhibits** in the memo: one establishes incidence or exposure and one establishes outcomes or distribution. A compact decision table may supplement them if the memo remains readable.

One partner submits for the pair.
Use `LastName-LastName-memo.pdf` and `LastName-LastName-research.zip`. The appendices verify the analysis; they are not extra pages for arguments omitted from the four-page memo.

The Required Four-Page Memo

A short decision document, not a slide deck or literature review

3. Each page has one job

Page	Job	What the reader must learn
1	Finding and scope	Name the decision-maker, population or sectors, period, baseline, and comparison. State the answer and confidence in the first substantive paragraph.
2	Economic mechanism	Map the shock through the relevant model to named factors, sectors, firms, workers, or households. Explain the countervailing force.
3	Evidence and incidence	Present the two principal exhibits. Establish magnitudes and a defensible incidence finding or ranking; distinguish exposure, association, and causal evidence.
4	Decision and limits	Address the strongest contrary evidence, distribution, uncertainty, sensitivity, policy implication, next verification step, and reversal condition.

Answer template. "For [defined groups] over [period], the evidence most strongly identifies [finding]. I have [confidence level] because [reason]. I recommend [action], and would revise it if [observable evidence]."

4. Evidence and writing standard

- Put the answer first and write in complete paragraphs. Use claim-based headings rather than "Background" or a generic list of pros and cons.
- Give every exhibit a claim-based title, labelled axes and units, geography and dates, source, and note explaining transformations.
- Recreate figures from downloaded data. Do not paste website screenshots.
- Label each important claim as an observed fact, model prediction, association, causal estimate, scenario, or recommendation.
- Interpret each exhibit in prose and state its limitation. End with a decision and a concrete next check, not only "more research is needed."

Ottawa-ready bar. A decision-maker could act tomorrow because the memo states the magnitude, affected group, mechanism, uncertainty, policy implication, and the evidence that would change the recommendation.

Required Welfare Ledger

Start from material living standards; broaden welfare only with measurable constructs

5. Build the analysis in five stages

1. **Define welfare before inspecting results.** Name the population, dimensions, units, preferred direction, time horizon, and the difference between an average gain and a universal claim.
2. **Measure the labour channel.** Define export-exposed, import-competing, downstream-input, and less-tradable worker groups using pre-period trade exposure. Track employment and earnings, not wages alone.
3. **Measure the consumption channel.** For each of the five income quintiles, construct 2023 category budget shares s_{gk} from household spending and match them transparently to price categories.
4. **Create at least four profiles.** Pair a worker group with an income-quintile basket and label the result an illustrative profile unless you have microdata that observe the joint cell.
5. **Add broader welfare only if useful.** Add at most two measurable non-income dimensions. For each profile, show the channel and evidence; do not hide conflicting dimensions inside an unsupported composite score.

For profile g, report the transparent approximation

$$\Delta \ln(\text{real resources}_g) \approx \Delta \ln(\text{labour income}_g) - \sum_k s_{gk} \, \Delta \ln p_k.$$

Show employment risk separately if the earnings series averages only employed workers. Do not add unlike historical episodes into one number as if they occurred at the same time.

If your design cannot identify a causal counterfactual price, use a transparent lower-trade-cost scenario in which documented trade-exposed retail categories fall by 2, 5, and 8 per cent. Call the result a *direct basket-price effect*, not exact compensating variation. For each profile, report the break-even earnings loss: the largest earnings decline that its consumption gain could offset. Reverse the sign for a higher-trade-cost scenario.

6. Required sensitivity and interpretation

Vary the price pass-through assumption and the wage/earnings response over a defensible range. Recalculate which profiles gain. State whether your aggregate finding is robust and whether any evidence can establish that *every* household gains. If welfare dimensions conflict, either defend alternative social weights or report that the overall ranking is ambiguous under unspecified weights. Ambiguity can be the correct conclusion.

Exporter wage premia can reflect selection; post-policy wage changes can reflect other shocks; CPI categories are not import prices; and group averages cannot prove that every household within the group gains. Match the confidence of the claim to the design.

Evidence Plan

Make the material foundation, any broader dimension, and distribution visible

7. Required exhibits and open-data starter pack

- **Exhibit 1 – labour channel:** trade exposure and employment or real earnings for the selected worker groups, with a pre-period, comparison group, and documented industry mapping.
- **Exhibit 2 – consumption channel:** budget shares by income quintile and the resulting cost-of-living effect under your central price scenario. Show which categories drive the difference. If you add a broader dimension, integrate it as a clearly labelled panel or compact dashboard.
- **Decision table:** for each profile, wage/earnings effect, employment risk, consumption-price effect, net partial-equilibrium result, any added welfare dimension, confidence, unresolved value judgment, omitted channels, and the policy implication.

Purpose	Official or original source
Household baskets	Statistics Canada Table 11-10-0223-01, household spending by income quintile
Consumer prices	Statistics Canada Table 18-10-0004-01, monthly CPI by product and geography
Employment and earnings	Statistics Canada Table 14-10-0221-01, seasonally adjusted monthly industry data
Trade exposure	Statistics Canada CIMT Web Application, monthly HS product and partner data
Recent price benchmark	Cavallo et al., Bank of Canada, The Price Impact of Canadian Retaliatory Tariffs
Historical Canadian adjustment	Trefler, The Long and Short of the Canada–U.S. Free Trade Agreement
Canadian welfare synthesis	Global Affairs Canada, State of Trade 2022: The benefits of free trade agreements
Broader welfare measures	Government of Canada, Quality of Life Framework; Statistics Canada, Quality of Life Hub

Confirm spending shares sum to the stated consumption denominator; document the SHS–CPI category crosswalk; distinguish price levels from inflation and nominal from real earnings; fix exposure weights before the shock; disclose revisions and missing cells; do not join group means as if they were linked microdata; and manually verify at least three prices, three spending cells, and three labour observations. For every added welfare dimension, verify construct, unit, direction, affected population, mechanism, horizon, and source. Do not combine dimensions without disclosed and defensible weights.

Codex Research Workflow

Use AI to accelerate auditable inquiry, not to outsource judgment

8. Use a dedicated project folder

```
project/
  README.md    sources.md    AI_LOG.md
  data/raw/    data/processed/    code/
  figures/    memo/    verification_report.md
```

Preserve downloaded files unchanged in `data/raw`. Generated files belong in `data/processed`. Never put passwords, tokens, private student data, or confidential information in a prompt.

9. A one-week research-assistant sequence

Day	Objective	Required practice
1	Work backward	Freeze the scope; define groups, baseline, comparison, mechanism, incidence rule, needed evidence, and possible refutations.
2	Acquire and audit	Download primary files; record definitions, units, classifications, geography, dates, access dates, and revision status.
3	Build measures	Write reproducible code; test joins, missingness, duplicates, totals, units, and identities. Manually verify three observations.
4	Build exhibits	Create the incidence finding and two exhibits; test an alternative denominator, window, or pass-through assumption.
5	Draft	Write the finding first, then mechanism, evidence, contrary case, decision, and reversal condition.
6	Red-team	Search for direction errors, unsupported causal language, fragile incidence findings, and evidence inconsistent with the preferred conclusion.
7	Reproduce	Run from raw data in a clean folder; regenerate every exhibit; audit the PDF, links, sources, AI log, and contributions.

10. Useful prompts

Plan: "Do not answer yet. Turn the assigned question into groups, mechanisms, measures, competing explanations, an incidence rule, and observations that could overturn each claim."
Data: "Inspect the documentation before coding. Build a data dictionary from actual columns. Flag classification joins, incompatible units or dates, revisions, missing values, and policy-direction errors. Invent nothing."
Audit: "Classify each material claim as fact, model prediction, association, causal estimate, scenario, or recommendation. Flag unsupported claims, denominator errors, fragile incidence findings, and exhibits that do not support the prose. Do not rewrite the memo yet."

AI output is not a source. You remain responsible for every number, citation, inference, and sentence. Record important retained output, errors, corrections, and each partner's contribution in the AI-use appendix.

Rubric and Submission Gate

The bar is a concise document that another researcher can audit

11. Assessment rubric – 100 points

Dimension	Weight	What level 5 demonstrates
Exact question and answer-first finding	15	Direct answer with defined groups, dates, baseline, comparison, confidence, and conditions.
Economic mechanism and incidence	20	Correct model mapped to named factors, sectors, regions, workers, consumers, or households.
Measurement and empirical design	20	Valid open-data measures; consistent units and classifications; descriptive and causal claims clearly separated.
Evidence, incidence finding, and exhibits	20	Two reproducible exhibits establish magnitudes and a defensible incidence finding or ranking that survives a meaningful sensitivity.
Judgment and Ottawa-ready usefulness	15	Contrary evidence, distribution, uncertainty, action, next check, and reversal condition are decision-relevant.
Professional memo, reproducibility, and AI transparency	10	Four readable pages, complete source trail, clean rerunnable code, preserved raw data, AI corrections, and contributions.

12. Levels and hard gates

Each dimension receives a level: **5 decision-ready and independently auditable**; **4 strong**; **3 correct and useful, but not yet decision-ready**; **2 partial or materially flawed**; **1 minimal**; or **0 missing or unsupported**. Row points equal weight × level/5.

No verified open data or reproducible code: overall maximum 50. An adjacent essay that does not answer the question: first row 0 and overall maximum 50. Theory named but not mapped to incidence: mechanism maximum level 2. Exposure or correlation presented as causal: design maximum level 2. No meaningful sensitivity: evidence maximum level 3. Fabricated or untraceable evidence: evidence and reproducibility receive 0. No distributional analysis or serious contrary case: judgment maximum level 2.

13. Verification and final upload

Every number traces to a dataset, variable, and raw observation; classifications, dates, currencies, prices, units, and denominators are consistent or disclosed; three observations and one official aggregate were checked by hand; raw data remain unchanged; exclusions and revisions are documented; all exhibits regenerate from a clean run; the required sensitivity is reported; and both partners can explain the memo, code, figures, and AI corrections.

Submit one memo PDF and one research ZIP through Canvas, then reopen both. Canvas is the authoritative deadline and submission record.

Final test. Could a skeptical policy reader reproduce the exhibits, understand exactly who gains and loses, see the best case against the finding, and know what evidence would change the decision?